

Thilakshan Elango

CSE Undergraduate – Full-Stack Developer – Software Engineer

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ABOUT

Computer Science and Engineering undergraduate at the University of Moratuwa with hands-on experience in building full-stack applications. Comfortable with modern web frameworks, backend systems, and cloud databases. Seeking a software engineering internship to deepen systems and product development skills

EDUCATION

University of Moratuwa

BSc Engineering (Honours) in Computer Science and Engineering

2024 – Present

Colombo, Sri Lanka

- Specialisation: Cyber Security

Kokuvil Hindu College

G.C.E. Advanced Level in Physical Science stream [Z-Score: 2.3314]

2014 – 2023

Jaffna, Sri Lanka

- District Rank: 13 | Island Rank: 411

EXPERIENCE

Rotaract Club of University of Moratuwa

2026

Volunteer – Logistics Team & Delegates Handling Team, Resume Center '26 Official Career Fair

- Volunteered at Resume Center '26 Official Career Fair organized by the Rotaract Club of the University of Moratuwa in collaboration with the Career Guidance Unit, supporting logistics and coordinating delegates for smooth event operations.
- Strengthened teamwork, communication, problem-solving, and event coordination skills while collaborating with volunteers.

Volunteer Physics Teacher

2023 – 2024

Volunteer Teaching

- Taught basic physics concepts to school students in a volunteer setting, preparing lesson materials and guiding students through problem-solving exercises.
- Improved students' conceptual understanding and exam readiness.

PROJECTS

AgriSenseNet – Smart Agriculture Platform 🌱 | Group Project

2026

- Built a precision agriculture platform using IoT sensors, real-time data pipelines, and ML models to automate irrigation decisions and crop monitoring for mixed-vegetable farming.
- The system collects sensor data (soil moisture, temperature, humidity, pressure, solar radiation) via ESP32 devices, streams through Apache Kafka, processes with ML models, and displays on a web/mobile dashboard.
- **Contribution:** Designed and implemented the complete data storage layer — PostgreSQL schema (7 tables) and InfluxDB time-series database on a Kubernetes-based cloud server. Built a real-time Kafka-based stream ingestion pipeline with sensor validation, and a daily batch pipeline computing Growing Degree Days (GDD) consumed by 6 downstream ML models. Wrote integration tests covering database connectivity, message validation, and end-to-end data flow.
- **Tech Stack:** Python, PostgreSQL, InfluxDB, Apache Kafka, Docker, GitHub, Kubernetes

Time Slot Booking Application 📅 | Ongoing

2026

- Developed a full-stack MERN appointment scheduling platform with JWT authentication, role-based access control (RBAC), booking CRUD, and real-time slot availability tracking.
- Engineered secure REST APIs with server-side validation, ownership-based authorization, refresh-token rotation, rate limiting, standardized error handling, and concurrency-safe booking using MongoDB unique compound indexing.
- **Tech Stack:** React.js, Node.js, Express.js, MongoDB Atlas, REST API

- MemSentry: Memory Forensics Platform  | Group Project | Ongoing2026
- Contributing to a portable memory-forensics platform that automates Volatility 3 analysis, correlates process, memory, thread, module, network, and YARA artifacts, and generates structured forensic reports.
 - Developing case management, SHA-256 integrity verification, provenance tracking, and automated reporting and testing to support reproducible forensic analysis.
 - Tech Stack:** Python, Volatility 3, YARA, SQLite, JSON, Git

- KandyPack Logistics Management System  | Group Project2025
- Developed a web-based logistics management system for order processing, transportation coordination, delivery operations, and management reporting.
 - Contribution:** Implemented the Reporting module, building the backend using Node.js and Express to generate six operational reports and enable PDF report generation.
 - Tech Stack:** React.js, Node.js, Express.js, SQL, Git, Docker

- Nano Processor Simulator | Digital Logic Design2024
- Designed and simulated a 4-bit nano-processor integrating an ALU, register bank, instruction decoder, program counter, ROM, multiplexers, and 7-segment display.
 - Implemented 2-way 3-bit, 2-way 4-bit, and 8-way 4-bit multiplexers, and contributed to FPGA deployment and system-level testing on the BASYS3 board.

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, C, C++, SQL
Frameworks & Libraries: React.js, Node.js, Express.js, Tailwind CSS
Tools & Databases: Git, GitHub, Docker, Linux, PostgreSQL, MongoDB, MySQL, InfluxDB, Apache Kafka, Postman, Jira, VS Code
Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming, Database Management, Software Engineering

ACHIEVEMENTS

- Top Performer** – CryptX 2.0 Capture The Flag Competition, ICT Society, Faculty of Technology, University of Sri Jayewardenepura (2026)
- Finalist** – UOJ Coders v5.0 Hackathon, organized by the Computer Society & Department of Computer Science, University of Jaffna (July 2026)
- Participant** – Battle of the Multiverse '26: Sri Lanka x Japan CTF Challenge (March 2026)

EXTRACURRICULAR & LEADERSHIP

- Volunteer** – CSE Careers Day 2026, University of Moratuwa (Logistics Team)
- Participant** – ENIGMA 2025 Virtual Hackathon, University of Moratuwa
- Player** – CSE HIT the Grounds, CSE'23 Batch, Department of Computer Science & Engineering
- Actor** – CSE Drama Fest, Department of Computer Science & Engineering

REFEREES

Prof. Dulani Meedeniya Senior Lecturer Department of Computer Science and Engineering University of Moratuwa Email: dulanim@cse.mrt.ac.lk	Dr. Adheesha Wijayasiri Senior Lecturer Department of Computer Science and Engineering University of Moratuwa Email: adeeshaw@cse.mrt.ac.lk
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